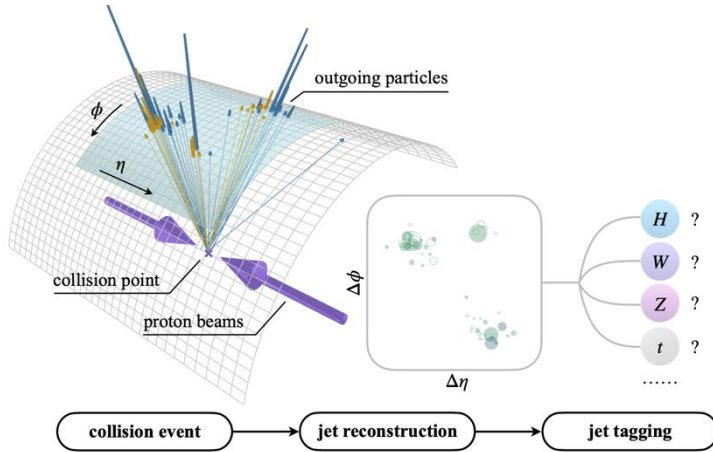


Transfer Learning for Jet Tagging at the LHC with Pretrained Foundation Models

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- LHC collides protons bunches ~ 40 M times/sec, these produce jets
- Training machine learning models from scratch to identify jets is expensive

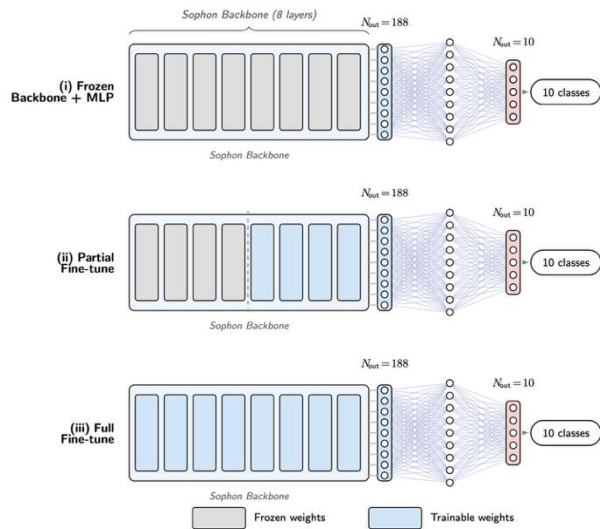
Particle Transformer (trained from scratch)

2.2M params · AUC 0.988

Ours (frozen Sophon + tiny head)

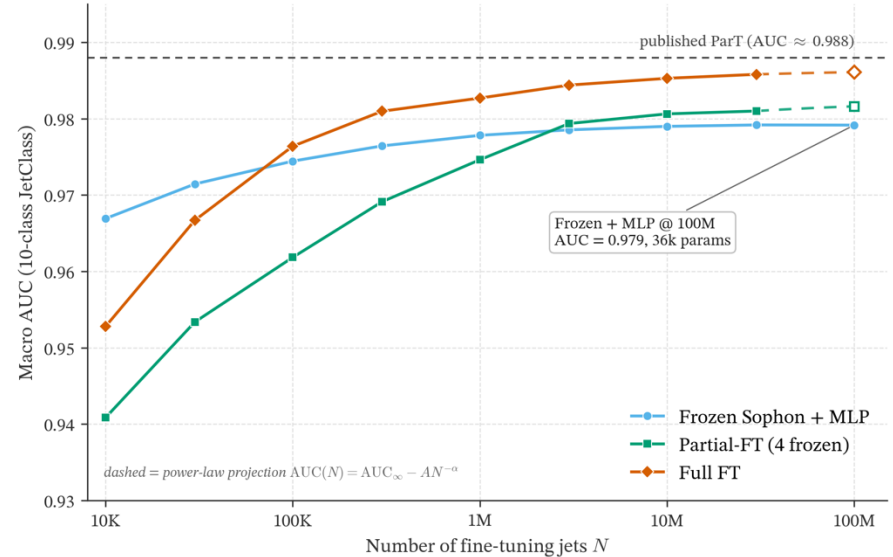
35K params · AUC 0.979

99% of the accuracy · 1.6% of the parameters



- Adapt Sophon (pretrained on 188 jet types) to a 10-class task
- 3 strategies: frozen head (35K), partial (1.4M), and full (2.2M)

Transfer-learning scaling: Sophon adaptations



- Scaling resembles power-law improvement curves
- Frozen + 35K head $\rightarrow$ 99% of ParT's AUC at 1.6% of the params